

Assignment 9: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, zoo, and trend packages
 - Set your ggplot theme

```
getwd()
```

```
## [1] "C:/SEA TEMP/EDE_Spring2026"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.1
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(zoo)
```

```
## Warning: package 'zoo' was built under R version 4.5.3
```

```
##  
## Attaching package: 'zoo'  
##  
## The following objects are masked from 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
library(trend)
```

```
## Warning: package 'trend' was built under R version 4.5.3
```

```
theme_set(theme_bw())
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1  
file_list <- list.files("Data/Raw/Ozone_TimeSeries",  
                        pattern = "*.csv",  
                        full.names = TRUE)  
  
GaringerOzone <- read.csv(file_list[1])  
  
for(i in 2:length(file_list)){  
  temp <- read.csv(file_list[i])  
  GaringerOzone <- rbind(GaringerOzone, temp)  
}  
  
dim(GaringerOzone)
```

```
## [1] 3589 20
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".

6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame `GaringerOzone`.

```
# 3
GaringerOzone$Date <- as.Date(GaringerOzone$Date, format = "%m/%d/%Y")

# 4
GaringerOzone <- GaringerOzone[, c("Date",
                                   "Daily.Max.8.hour.Ozone.Concentration",
                                   "DAILY_AQI_VALUE")]

# 5
Days <- data.frame(Date = seq(as.Date("2010-01-01"),
                              as.Date("2019-12-31"),
                              by = "day"))

# 6
GaringerOzone <- left_join(Days, GaringerOzone, by = "Date")

dim(GaringerOzone)

## [1] 3652    3
```

```
head(GaringerOzone)
```

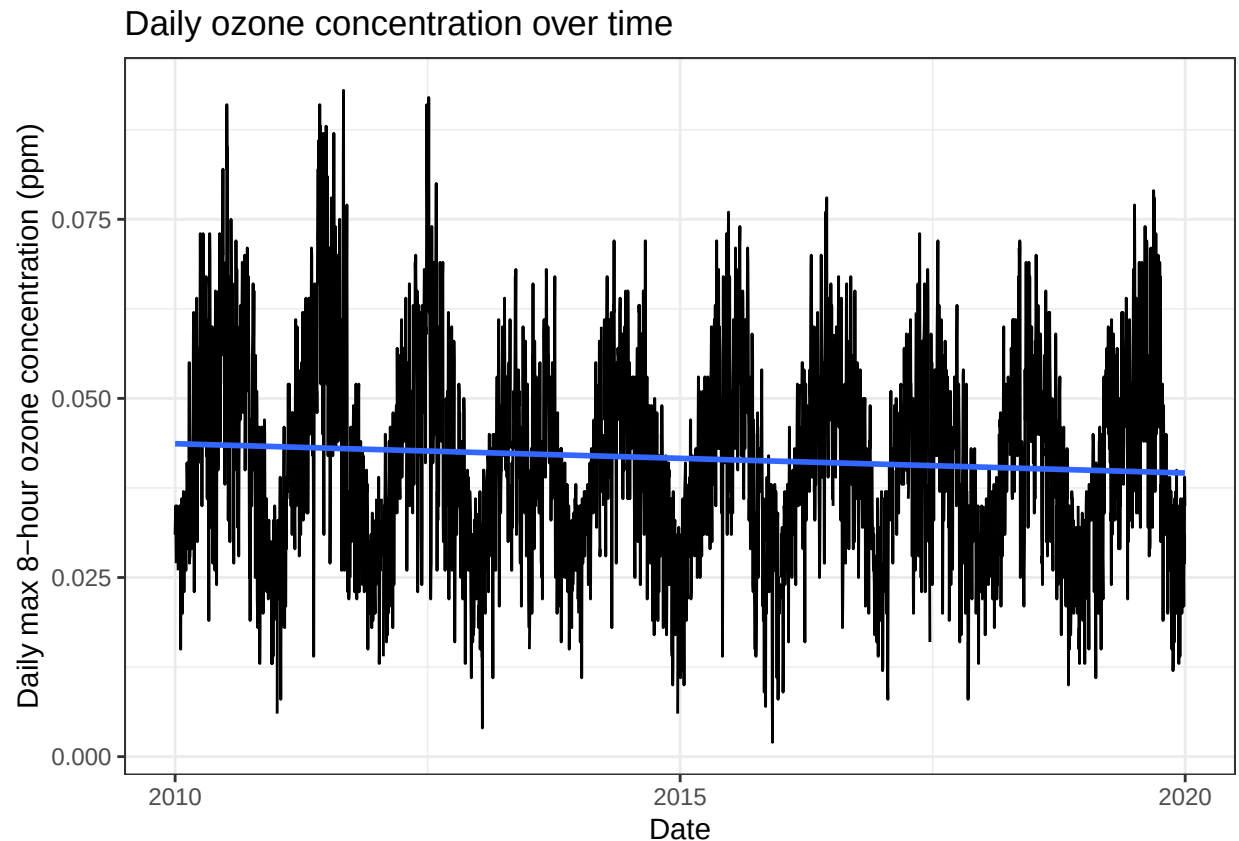
```
##      Date Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
## 1 2010-01-01                        0.031                29
## 2 2010-01-02                        0.033                31
## 3 2010-01-03                        0.035                32
## 4 2010-01-04                        0.031                29
## 5 2010-01-05                        0.027                25
## 6 2010-01-06                        NA                 NA
```

Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
ggplot(GaringerOzone,
       aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line(na.rm = TRUE) +
  geom_smooth(method = "lm", se = FALSE, na.rm = TRUE) +
  labs(x = "Date",
       y = "Daily max 8-hour ozone concentration (ppm)",
       title = "Daily ozone concentration over time")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Answer: The plot shows strong seasonal variation in ozone concentration over time. The smoothed line suggests a slight downward trend over the 2010s, so ozone concentrations may have decreased somewhat at this station.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station? 8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone$Ozone_interp <- na.approx(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,
                                         na.rm = FALSE)
```

Answer: Linear interpolation is appropriate because ozone concentration changes continuously over time. Piecewise constant interpolation would create unrealistic flat sections, while spline interpolation could make the data too smooth and less realistic.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new `Date` column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```

#9
GaringerOzone$Year <- format(GaringerOzone$Date, "%Y")
GaringerOzone$Month <- format(GaringerOzone$Date, "%m")

GaringerOzone.monthly <- GaringerOzone %>%
  group_by(Year, Month) %>%
  summarise(Mean_Ozone = mean(Ozone_interp, na.rm = TRUE),
            .groups = "drop")

GaringerOzone.monthly$Date <- as.Date(paste(GaringerOzone.monthly$Year,
                                           GaringerOzone.monthly$Month,
                                           "01",
                                           sep = "-"))

head(GaringerOzone.monthly)

```

```

## # A tibble: 6 x 4
##   Year  Month Mean_Ozone Date
##   <chr> <chr>      <dbl> <date>
## 1 2010   01         0.0305 2010-01-01
## 2 2010   02         0.0345 2010-02-01
## 3 2010   03         0.0446 2010-03-01
## 4 2010   04         0.0556 2010-04-01
## 5 2010   05         0.0466 2010-05-01
## 6 2010   06         0.0576 2010-06-01

```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```

#10
GaringerOzone.daily.ts <- ts(GaringerOzone$Ozone_interp,
                             start = c(2010, 1),
                             frequency = 365)

GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Mean_Ozone,
                               start = c(2010, 1),
                               frequency = 12)

```

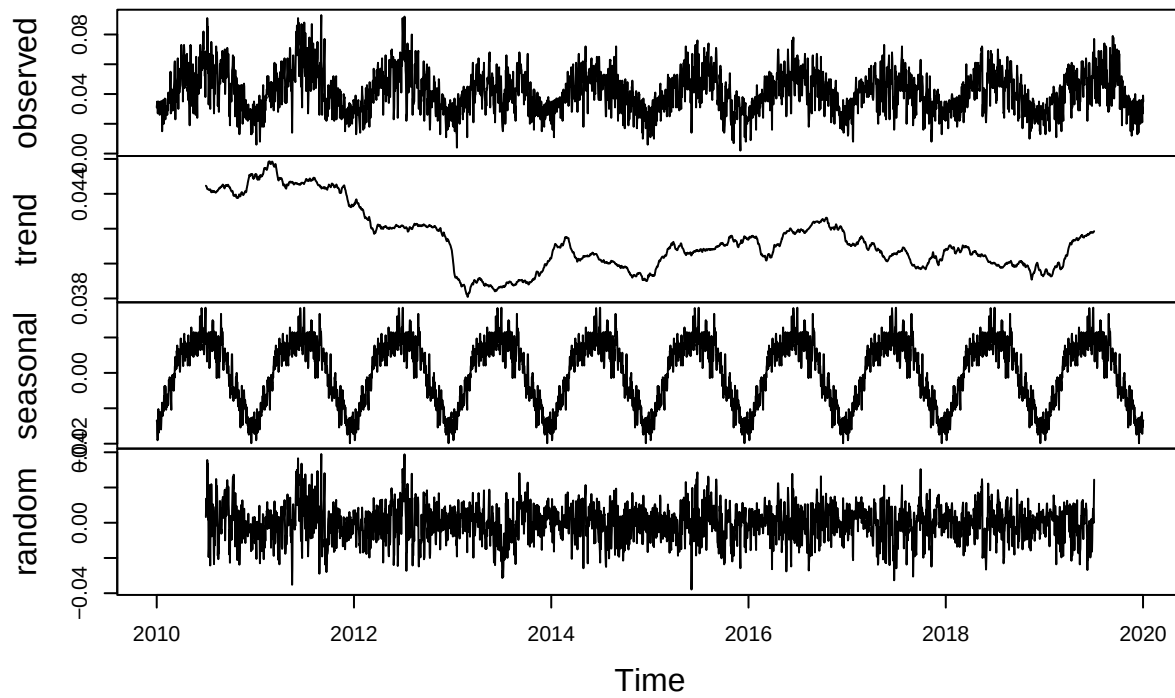
11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```

#11
daily.decomp <- decompose(GaringerOzone.daily.ts)
plot(daily.decomp)

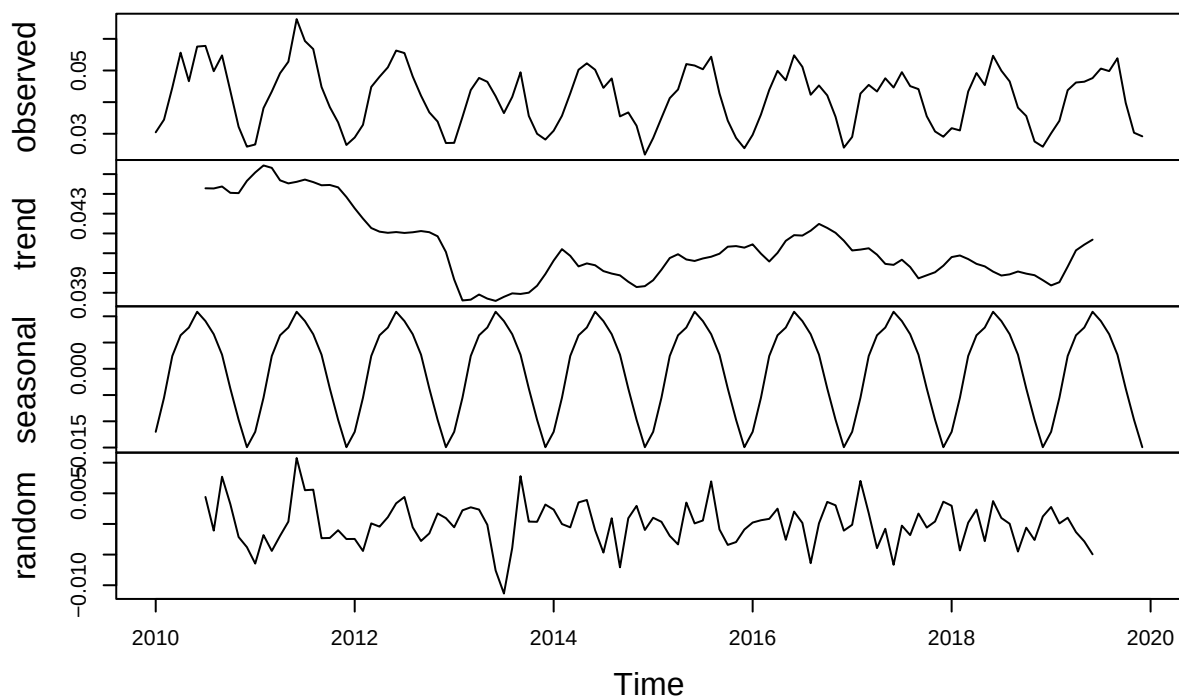
```

Decomposition of additive time series



```
monthly.decomp <- decompose(GaringerOzone.monthly.ts)
plot(monthly.decomp)
```

Decomposition of additive time series



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

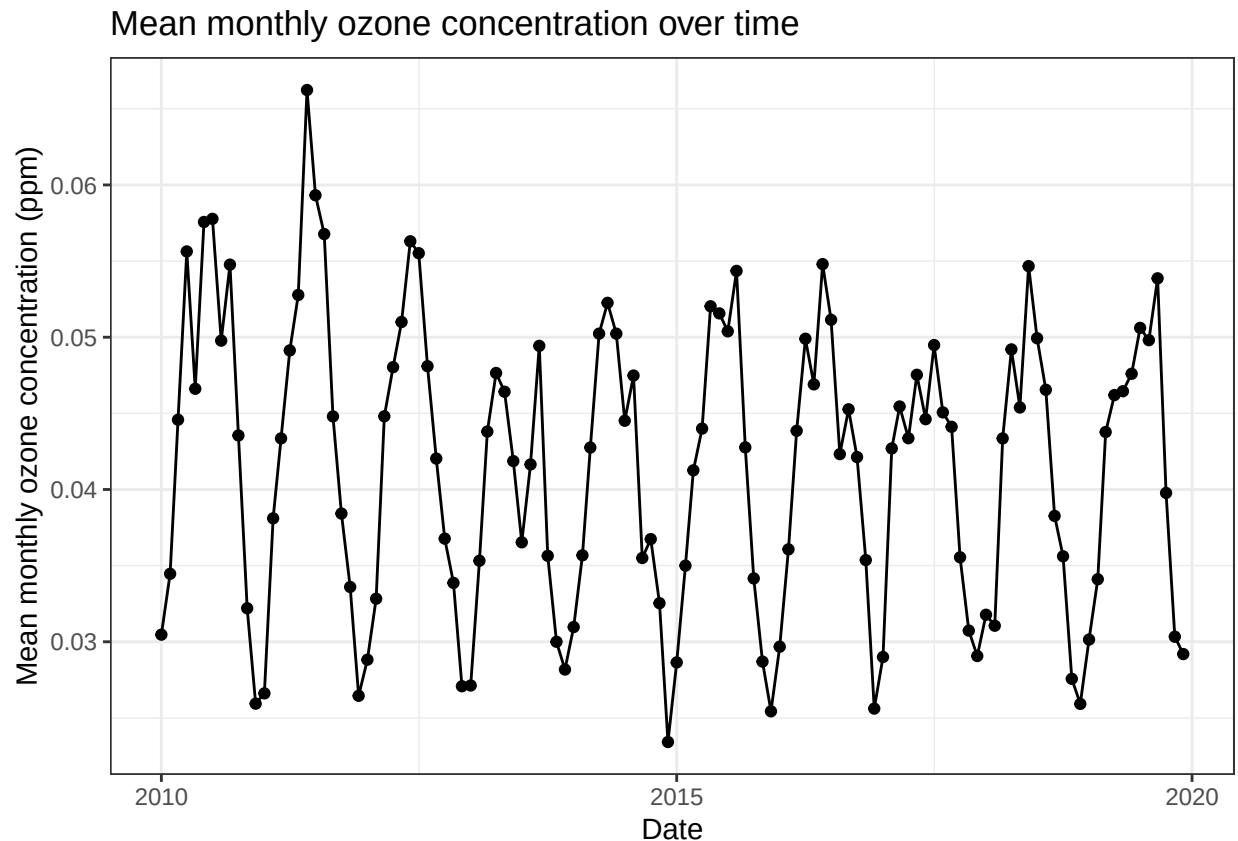
```
#12
smk_result <- smk.test(GaringerOzone.monthly.ts)
smk_result

##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -1.963, p-value = 0.04965
## alternative hypothesis: true S is not equal to 0
## sample estimates:
## S varS
## -77 1499
```

Answer: The seasonal Mann-Kendall test is best because ozone concentrations change by season and that pattern repeats each year. It is used to test the long-term trend while accounting for seasonality.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly, aes(x = Date, y = Mean_Ozone)) +
  geom_point() +
  geom_line() +
  labs(x = "Date",
       y = "Mean monthly ozone concentration (ppm)",
       title = "Mean monthly ozone concentration over time")
```



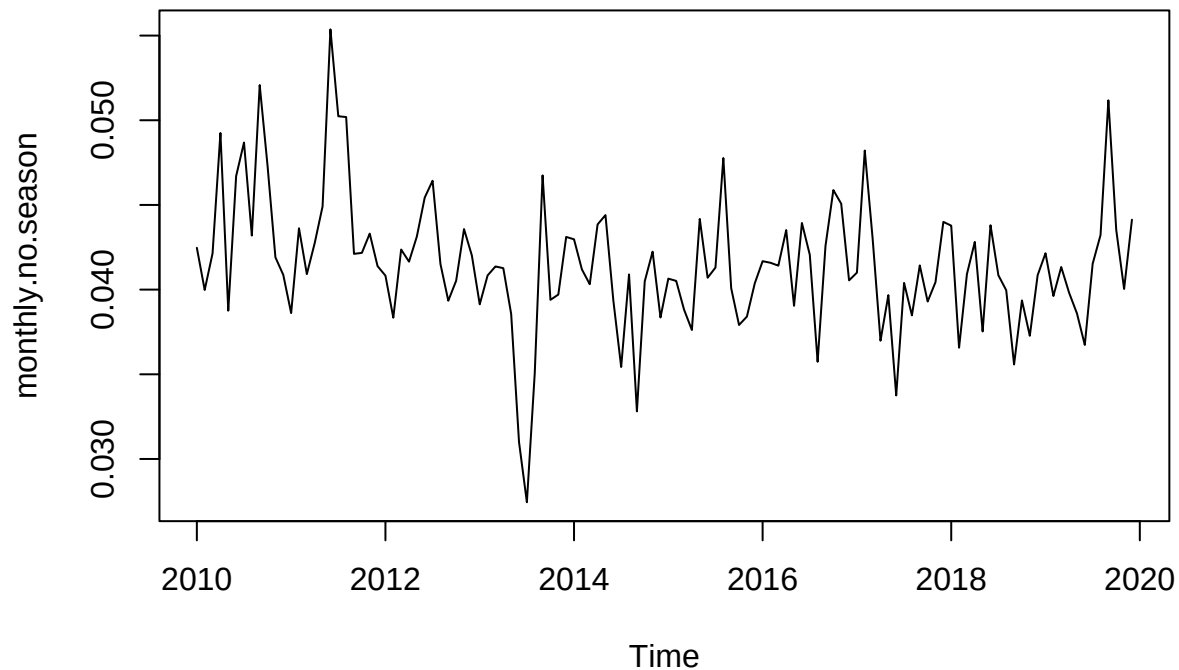
14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Ozone concentrations appear to have decreased slightly over the 2010s at this station. The Seasonal Mann-Kendall test indicates that this trend is statistically significant, even though the monthly data still show strong seasonal variation ($z = -1.963$, $p\text{-value} = 0.04965$).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
# 15
monthly.no.season <- GaringerOzone.monthly.ts - monthly.decomp$seasonal

plot(monthly.no.season)
```



```
# 16
mk_result <- mk.test(monthly.no.season)
mk_result

##
## Mann-Kendall trend test
##
## data: monthly.no.season
## z = -2.6039, n = 120, p-value = 0.009216
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## -1.149000e+03  1.943657e+05 -1.609356e-01
```

Answer: The deseasonalized monthly ozone series still shows a significant downward trend over time. This agrees with the Seasonal Mann-Kendall test, because both tests found a significant decrease, although the Mann-Kendall test on the deseasonalized series gave stronger evidence of a trend (MK: $z = -2.6039$, $p\text{-value} = 0.009216$; Seasonal MK: $z = -1.963$, $p\text{-value} = 0.04965$).